CpE 3110/CS3803 Homework Assignment #5 Maximum – 100pts

You must always show or explain your work in a neat and orderly format. You are encouraged to discuss ideas with other students and consult references, but your work must be your own.

1. Suppose you are given two different LEGv8 processors, P1 and P2, namely. P1 is a single-cycle processor with clock cycle time of 800ps. P2 is a **7-stage** pipelined processor with the clock cycle time of 200ps. If you are given a program with 1000 instructions without any pipeline hazards, calculate the execution times on P1 and P2. Then, calculate the speedup (i.e., the ratio of execution times of P1 and P2) [25pts].

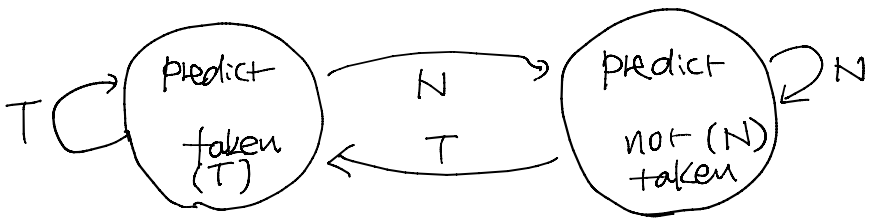
2. Suppose you are given a 5-stage pipelined LEGv8 processor with both EXE and MEM forwarding paths added. Consider the following instruction sequence [30pts]:

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ADD X9, X9, X10
ADD X9, X9, X11
LDUR X9, [X9, #0]
ADD X9, X9, X12
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a) Clearly identify all data hazards.

b) For each identified data hazard, which forwarding path should be used? Do you need to insert any stall clock cycle(s)? Justify your answer.

3. Suppose you are given a pipelined LEGv8 processor with built-in 1-bit dynamic predictor as shown below. Calculate the branch prediction accuracy (= # of correct predictions / # of total predictions) of the following branch outcome sequence. Assume that the initial state is "Predict taken" [30pts].



Branch outcome sequence (T means "branch-taken" and N means "branch-not-taken"):

T, T, N, N, N, T, N, T, N, T, T, N, N

4. For the branch outcome sequence given in question 3, calculate the branch prediction accuracy of the static branch prediction strategies including "Always predict taken" and "Always predict not taken". Among three branch prediction

strategies considered in questions 3 and 4, which one has the highest branch prediction accuracy for the given branch outcome sequence? Justify your answer [15pts].